AI SYNERGY HACKATHON 2026 / ROUND 1

One Complaint. Five Departments.

Case Investigation / Problem–Solution Brief

TRACK Governance	**CASE** 07
WORKING PROJECT NAME CivicTrace	**STATUS** Provisional problem framing

Our starting point

A complaint should remain understandable as it moves through a public system: what was reported, which office is accountable for the next action, what has changed, and why. The case raises questions about whether that trail remains clear across channels and departments. We are investigating that possibility rather than treating every reassignment as a failure.

OUR PROVISIONAL FOCUS

Test whether unclear ownership or incomplete case histories make municipal complaints difficult to track and independently reconstruct.

Team: 3 third-year CSE students • Current stage: investigation and prototype planning

Evidence note: the case booklet's 23% reassignment figure is simulated. It is not presented as a real municipal statistic.

01 | Case and stakeholders

Selected case

Case 07 — Governance: “One Complaint, Five Departments.” The scenario describes complaints arriving through portal, phone, social media and in-person channels; partially separate workflows; uncertainty about the responsible department; generic “in process” statuses; multilingual and multimodal submissions; and a need for an auditable record of routing and prioritisation decisions.

Stakeholders

STAKEHOLDER	WHAT THEY NEED	QUESTION TO VALIDATE
Residents / complainants	Clear status, next step and route to correct information.	Can they identify who owns the complaint and what happened last?
Grievance / intake staff	Usable history of submissions, assignments and actions.	Do they reconcile information across records or channels?
Department handlers	Clear responsibility and relevant context at handoff.	Can they see why the case reached them and what has been done?
Ward officials / administrators	Visibility into pending cases and responsibility for next action.	Can responsibility be identified for multi-department cases?
Auditors / authorised reviewers	A reconstructable decision history, including reasons and overrides.	Can a reviewer explain routing and status from the record alone?

Constraints we will not compromise

- No complaint is automatically rejected or closed solely on an AI score.
- Human review remains responsible for consequential routing, priority and closure decisions.
- The original submission and later corrections remain traceable; duplicate detection must not erase distinct reports.
- Residents must have an accessible way to correct information or contest a decision.

02 | Evidence, assumptions and alternatives

What the case tells us — and what it does not

CASE SIGNAL	STATUS	WHAT WE CAN SAY
Multiple intake channels and partially separate workflows	**Stated in case**	A fragmentation risk is worth investigating; the extent of record disconnection is not established.
Residents often do not know the responsible department; generic statuses can persist	**Stated in case**	The scenario describes an unclear resident experience; it does not prove internal ownership records are absent.
23% reassigned at least once	**Simulated data**	A prompt for investigation, not a real-world rate and not proof of incorrect routing or delay.
Low-frequency complaints may be safety-critical	**Risk flagged**	Safety must be considered in testing; no actual missed safety case is established by this signal alone.
Existing audit trails may be insufficient	**Unknown**	Auditability is required, but the case does not prove current records fail this requirement.

Alternative problem statements considered

A. Avoidable complaint-routing delays	A clear, testable mechanism. Not selected as primary yet: reassignment may be legitimate, and the case does not show that avoidable transfers cause delay.
B. Fragmented ownership and traceable case records	Selected for validation because it connects directly to multiple workflows, responsibility uncertainty and the audit requirement. Still a hypothesis: existing systems may already preserve adequate records.
C. Safety-critical prioritisation and auditability	Potentially severe, but no real missed escalation or volume-driven prioritisation failure has been established. Retained as a safety constraint and alternative if evidence emerges.

Why B for now: it is the most feasible direction to validate through workflow inspection and case-history reconstruction. We are not claiming it is the proven root cause of routing delays or safety failures. If records already show clear ownership and a complete decision trail, we will reconsider the choice.

03 | Provisional problem and proposed solution

Problem statement

When municipal complaints arrive through multiple channels or move between departments, the current workflow may not consistently make the complaint's accountable office, assignment history, status changes and reasons for transfer easy to trace. This may make it difficult for residents and authorised reviewers to determine who is responsible for the next action and how the complaint has been handled. We will validate whether this gap exists before claiming that it causes delays or harm.

Proposed concept: CivicTrace

A lightweight, human-supervised case-history workspace that keeps the original complaint linked to subsequent assignments, actions, status changes and review decisions. The prototype will demonstrate the record and accountability workflow—not claim to replace a municipality's existing system.

CAPABILITY	WHAT THE PROTOTYPE DOES	HUMAN CONTROL
One traceable case ID	Links the original submission and later events to the same case history.	Staff confirm links; distinct reports remain preserved.
Ownership / handoff log	Records current office, prior assignments, timestamps and transfer reasons.	A handler confirms ownership and why a transfer is needed.
Readable status timeline	Shows last meaningful action, current stage and next responsible action.	Status reflects recorded actions, not invented progress claims.
AI-assisted intake demo	Suggests structured fields or a category; may flag a possible duplicate for review.	Suggestions are not decisions; no automatic rejection, closure or irreversible merge.
Review / correction trail	Stores corrections, overrides and reasons beside the original entry.	Authorised staff decide; residents can request correction or appeal.

Scope for a three-person team

- **Member 1 — workflow and evidence:** document the current process, assumptions and test scenarios.
- **Member 2 — product and interface:** build the case view, ownership timeline, status and correction workflow.
- **Member 3 — data and evaluation:** prepare synthetic cases, run reconstruction tests and record errors.

This is a small demonstrator. It will not connect to live municipal systems, make final priority decisions or claim to reduce actual resolution time without field evaluation.

04 | Validation plan and success measures

Smallest study before locking the problem

- Choose one target municipality or label the exercise as a generic workflow study; the case does not identify a source municipality.
- Review public workflow documents and, if permitted, request 30–50 anonymised case histories covering different channels and some transferred cases.
- Have two reviewers independently identify the current owner, assignment sequence, latest meaningful action and transfer reasons. Record missing fields and disagreements.
- If records are unavailable, conduct a few consent-based workflow walkthroughs and use synthetic cases for prototype testing. This tests feasibility, not real-world prevalence.
- Retain B only if a repeatable, consequential ownership or traceability gap is observed. Otherwise reconsider A or C against the evidence.

Proposed prototype acceptance criteria

1	**Persistent case history:** all 30 synthetic test cases retain a stable ID, assignment history and identifiable accountable office (or an explicitly assigned coordinating office).
2	**Independent reconstruction:** two reviewers correctly identify owner, transfer sequence, latest action and current status in at least 90% of 20 synthetic case-history assessments.
3	**Safety and correction guardrail:** no AI-only rejection or closure; all seeded safety scenarios requiring review are surfaced for human review; corrections preserve the original entry and decision trail.

Metrics

BASELINE	OUTCOME	HARM / EXCLUSION
Case-history reconstruction rate Share of sample cases for which reviewers can correctly reconstruct owner, assignments and latest meaningful action.	**Change in reconstruction rate** Percentage-point difference using the prototype on the same cases and scoring rubric; also record median review time.	**Missed safety-review rate** Share of seeded safety scenarios not surfaced for human review, reported overall and by English, Hindi and mixed-language input.

A synthetic test can show whether the prototype behaves as intended; it cannot prove that a real municipality has this problem or that the prototype improves civic outcomes.

05 | Risks, accountability and next decisions

Risks we will test rather than hide

RISK	HOW WE LIMIT IT
False duplicate match	Suggest links only; preserve each report; require human confirmation before linking records.
Wrong department suggestion	Show the reason and uncertainty; allow correction; record who confirmed assignment.
Language or format exclusion	Test English, Hindi and mixed-language examples; do not assume photo/voice interpretation is reliable without evaluation.
False sense of progress	Display only status and actions supported by recorded events.
Safety signal missed	Use conservative human-review flags in the prototype; AI cannot close or reject a complaint on its own.
Citizen information exposed	Use synthetic data for development; use anonymised records only with permission and collect only necessary fields.

Next decisions

- **Before final submission:** identify the target workflow and add only sources the team has opened and checked directly.
- **During testing:** preserve failed cases and reviewer disagreements; do not report only successful examples.
- **Before locking the problem:** state what supports B, what remains unknown, and what would make the team switch to A or C.

Why this is worth testing

A traceable complaint history is a practical unit of improvement: reviewers can test whether they can identify the accountable office and reconstruct what happened. The proposal is deliberately narrower than “fix municipal grievance redressal.” It can produce a defensible finding even if the hypothesis turns out to be wrong.

Evidence and reference note

Primary basis: AI Synergy Hackathon 2026, Case 07 — “One Complaint, Five Departments” case scenario, the team’s three investigation reports and consolidated evidence matrix. The reassignment figure is simulated. External reports mentioned during preliminary research are not used here as verified proof; add them only after checking the original source, date, exact passage and limitations.

WORKING CONCLUSION

CivicTrace is a proposed prototype for testing complaint ownership and case-history traceability. The selected problem remains provisional until workflow evidence is reviewed.

External Evidence Notes

Case 07 — “One Complaint, Five Departments” | Context sources and limits

How these sources are used: They provide context for investigating grievance redressal and accountability. They do not prove that the selected ownership/traceability failure exists in a specific municipality, nor that it causes delays. The case booklet’s 23% reassignment figure remains simulated competition data.

1. Aggarwal, Veena; Krithika, P. R.; Garg, Deepika (2010). “Strengthening municipal governance for effective consumer grievance redressal.” International Journal of Regulation and Governance, 10(2), 63–75.

Direct source: https://doi.org/10.3233/IJR-120095

Relevant finding: Finding relevant to our investigation: an examination of IT-enabled grievance redressal in a few Indian cities argues that technology alone is insufficient without accountability, transparency, community involvement and capacity.

Limitation: Limit: the article is older and limited to a few cities; it does not demonstrate a current traceability gap or prove transfer-caused delay in our target workflow.

2. Department of Administrative Reforms and Public Grievances (DARPG). Comprehensive Guidelines for Handling the Public Grievances, Office Memorandum S-15/27/2024-PG Division (2024).

Direct source: https://www.darpg.gov.in/sites/default/files/Comprehensive_guidelines_for_handling_the_Public_Grievances.pdf

Relevant finding: Relevant guidance includes transferring a grievance to the appropriate authority rather than closing it merely because it does not pertain to the receiving office; recording action-taken details and valid reasons for closure; and retaining an appeal process.

Limitation: Limit: this guidance concerns CPGRAMS and central-government grievance handling. It is not proof of how any municipal system operates or whether it follows the guidance.

3. Comptroller and Auditor General of India (CAG). Audit Report on Panchayati Raj Institutions and Urban Local Bodies, Uttar Pradesh, Report No. 2 of 2015, year ended 31 March 2015, Section 5.15.

Direct source: https://cag.gov.in/uploads/download_audit_report/2015/Report_No2_of_2015_-_Panchayati_Raj_Institutions_and_Urban_Local_Bodies_Government_of_Uttar_Pradesh.pdf

Relevant finding: The report describes grievance-redressal and complaint-register/monitoring deficiencies in specific test-checked local bodies, including unrecorded complaints and missing monitoring/disposal records.

Limitation: Limit: historical and limited to the audited entities/period. It does not establish current nationwide prevalence, an AI-related failure, or the cause of a specific delay.

4. Srivatsa, Vidula et al. (2026). “Reimagining Digital Grievance Redressal Systems for Better Participatory Governance in India.” DIS ’26 Companion, ACM.

Direct source: https://doi.org/10.1145/3802974.3810897

Relevant finding: The extended abstract reports a survey of 157 urban residents and in-depth interviews, and discusses institutional silence and siloed teams as concerns in digital grievance redressal.

Limitation: Limit: an extended abstract; sampling and representativeness need careful review. Its findings do not prove the selected ownership/record gap or its causal impact in a particular municipality.

Appropriate conclusion for Round 1: Independent literature and official guidance justify examining accountability, record quality and grievance handling. They do not replace validation of a selected municipal workflow. The next step is to inspect actual process documentation and, if permission is granted, a small anonymised sample of case histories.